

Scalar idempotents and projective cancellation from a torsion-free characteristic-two construction

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Abstract

The October 4 manuscript of OpenAI constructs a finitely presented torsion-free group G , with a finite two-dimensional classifying complex, and $a, b, c \in \mathbb{F}_2[G]$ such that $ab = 1$, $ac = 0$, and $c \neq 0$. Classical ring algebra then makes $e = 1 - ba$ a nontrivial scalar idempotent. For $R = \mathbb{F}_2[G]$, the right ideal $P = eR$ is nonzero, cyclic and projective, with $R \cong R \oplus P$ and $[P] = 0$ in $K_0(R)$. The same example works over every characteristic-two field. We give the complete consequence proof, an existential presentation recipe, and the scope of the source audit. The result and short proof were already disclosed in the author's public research triage. This record adds verification and exposition, claims no new mathematical discovery, and is withheld from a duplicate novelty preprint deposit.

1 Exact statement and attribution

For a field K and a torsion-free group G , the scalar form of Kaplansky's idempotent conjecture asks whether every $e \in K[G]$ with $e^2 = e$ belongs to $\{0, 1\}$; see [2, Problem 1(c)]. Its positive-characteristic form quantifies over all fields of positive characteristic. The example below contradicts its characteristic-two specialization, even for finitely presented groups with a finite two-dimensional classifying complex.

The pivotal input is precisely [1, Theorem 1.1], at repository commit [adc7f124](#) (full hash in the source manifest): there exist such a group G and $a, b, c \in \mathbb{F}_2[G]$ with

$$ab = 1, \quad ac = 0, \quad c \neq 0. \tag{1}$$

The group construction and its new failure of direct finiteness belong to that source. The reduction from failure of direct finiteness to a nontrivial idempotent is classical and explicitly appears in [3, Proposition `kaplansky_relations`]. The module calculation below is the corresponding standard splitting; compare the interpretation of direct finiteness of the regular right module in [4, p. 9].

Theorem 1. *There exists a finitely presented torsion-free group G admitting a finite two-dimensional classifying complex and a scalar idempotent $e \in R = \mathbb{F}_2[G]$ with $e \neq 0, 1$. The right R -module $P = eR$ is nonzero, cyclic and projective, and*

$$R_R \cong R_R \oplus P, \quad [P] = 0 \quad \text{in } K_0(R).$$

For every characteristic-two field K , the image of the same e in $K[G]$ remains an idempotent other than 0 and 1.

The algebraic proof is self-contained in the next section. The existence of the group is supplied by the cited input, whose actual probabilistic and topological arguments were separately audited as recorded in Section 3. Neither a scalar idempotent in odd characteristic or characteristic zero, nor a projection in a reduced group C^* -algebra, is asserted.

2 Ring and right-module proof

Proposition 2. *Let R be a nonzero unital ring and suppose (1) holds in R . Then $e = 1 - ba$ is an idempotent other than 0 and 1, and $P = eR$ has all the right-module and K_0 properties in Theorem 1. No characteristic assumption is needed for this proposition.*

Proof. Put $f = ba$. Then $f^2 = b(ab)a = f$, so $e^2 = (1 - f)^2 = 1 - 2f + f^2 = 1 - f = e$. Also $ec = c - b(ac) = c \neq 0$, whence $e \neq 0$. If $e = 1$, then $ba = 0$, and $a = a(ba) = 0$ contradicts $ab = 1$ in a nonzero ring. Thus $e \neq 1$.

The complementary idempotents f and e satisfy $fe = ef = 0$ and $f + e = 1$. Therefore $R = fR + eR$. If $x = fr = es$, then $x = fx = fes = 0$, giving the direct sum

$$R_R = baR \oplus eR. \quad (2)$$

Since $baR \subseteq bR$ and $b = bab$, the reverse inclusion holds as well: $baR = bR$. Left multiplication by b is a right R -linear isomorphism $R_R \rightarrow bR$, with inverse left multiplication by a . Indeed $abr = r$ for every r , and $ba(br) = br$ for every $br \in bR$.

Consequently (2) identifies R_R with $R_R \oplus P$. For a fully explicit pair of inverse maps, use

$$\Phi(r) = (ar, er), \quad \Psi(u, p) = bu + p \quad (r, u \in R, p \in eR).$$

The identities $ae = eb = 0$, $ep = p$, and $ab = 1$ give $\Psi\Phi(r) = bar + er = r$ and $\Phi\Psi(u, p) = (u, p)$. They also show $\ker(r \mapsto ar) = eR$; in particular $c \in P$.

The module P is a direct summand of the free right module R_R , hence projective. It is cyclic, generated by e , and nonzero because it contains $e = e \cdot 1 \neq 0$. By definition, $K_0(R)$ is the group completion of the direct-sum monoid of finitely generated projective right modules [5, Chapter II, Section 2]. Hence the isomorphism implies $[R] = [R] + [P]$ and therefore $[P] = 0$ in that abelian group. It does not imply $P = 0$. Rather, $R \oplus 0 \cong R \oplus P$ with $0 \not\cong P$ exhibits failure of projective cancellation. $\square$

Proof of Theorem 1. Apply Proposition 2 to the group and triple of [1, Theorem 1.1]. If K is any characteristic-two field, the unique unital embedding $\mathbb{F}_2 \hookrightarrow K$ induces

$$\mathbb{F}_2[G] \longrightarrow K[G], \quad \sum_g \lambda_g g \longmapsto \sum_g \lambda_g g.$$

This map is multiplicative and injective because the elements of G form a basis over each coefficient field. It preserves $e^2 = e$ and cannot send e or $e - 1$ to zero. Thus the same finite-support scalar idempotent remains nontrivial over every such field. Applying the ring proposition there also gives $eK[G]$ with the corresponding absorption and zero K_0 class. $\square$

Vanishing of this one class does not assert $K_0(R) = 0$. In fact the scalar inclusion $\mathbb{F}_2 \rightarrow R$ and augmentation $R \rightarrow \mathbb{F}_2$ split off $K_0(\mathbb{F}_2) = \mathbb{Z}$ from $K_0(R)$. The characteristic-two field embedding supplies no coefficient map to a field of odd characteristic or characteristic zero.

3 Existence construction and explicitness

The source uses the projective plane over $\mathbb{F}_{128}$, with $v = 16513$ points and lines, and a signed alphabet of $v + 7 = 16520$ letters. The seven extras are paired with seven ordinary letters; the remaining ordinary letters are paired amongst themselves. Thus its rose has 8260 generators. For sufficiently large $m \equiv 1 \pmod{4}$, take m vertices per line class on side A and $m - 1$ on side B . Each ordinary outgoing-label set is the corresponding 129-point line. One distinguished A vertex has all seven extras. The other extra sets are empty or complements of Fano-plane lines. Each of those seven four-element complements occurs at $(129m - 1)/4$ vertices of A and $129(m - 1)/4$ vertices of B , distributed evenly amongst line classes. This balances inverse-label slots and imposes the exact intersection parities needed by the source’s Proposition 2.1.

Independent random bijections pair inverse slots on each side. Sections 3–5 of [1] prove positive probability of an outcome with logarithmic girth, bounded component diameter, and no reduced spherical arrangement. In Section 6 the cone-picture surgeries give both asphericity and protection of the designated roots. Attaching cones on all graph components to the rose yields the finite two-dimensional classifying complex and its torsion-free fundamental group.

For clarity, a finite presentation is available *after an outcome has been selected*. In each component Λ choose a spanning tree and root q_Λ , and let w_x label the tree path from q_Λ to x . For each non-tree edge with chosen orientation $x \rightarrow y$ and letter t , impose the relator $w_x t w_y^{-1}$. These relators normally generate the labels of all graph loops, exactly as required by the cone attachment. Choose the designated roots in their respective components A', B' . Then the source supplies

$$a = \sum_{x \in A'} w_x, \quad b = \sum_{x \in A'} w_x^{-1}, \quad c = \sum_{y \in B'} w_y^{-1}$$

in the presented group algebra. A simultaneous-label step makes each product contribution constant on a component of the product graph. The handshake lemma gives cancellation in characteristic two: all degrees are odd on $A' \times B'$, while the single even-degree exception (x_A, x_A) on $A' \times A'$ leaves contribution 1. Root protection ensures that only $y = x_B$ contributes to the identity coefficient of c , so that coefficient equals 1.

The parameter m and the matching outcome are existential in this source. No numerical graph, relator list, explicit reduced support of e , or multiplication certificate is supplied here. The tree recipe and the source’s fully specified probability model establish existence; they are not a claim that a finite search certificate was obtained.

Our dependency audit checked the repeated-edge issue by reconstructing the multiplicity-stage first moments and the nearly aligned block grid. Different stages need not be independent: a realization must survive each separate first-moment bound. The small unpaired fraction is fixed before the path-number, length, and comparison bounds. Planar extraction then applies one fixed triple of bounds to arrangements of every size. The topology audit checked all four same-edge band surgeries, including preservation of graph endpoints in rooted disk pictures and essentiality after splitting a sphere. These checks are recorded with exact source hashes and limitations in the accompanying research records.

The family-197 Lean documentation names earlier torsion-containing examples. Its actual finitely presented declaration includes an element of odd prime order. We did not use that formalization to verify the October 4 torsion-free theorem and make no claim that this consequence note is formally verified. An isolated rebuild of the pinned source compiled its 31-page manuscript; compilation is a document check, not proof verification. Automated audits are also not human peer review.

4 Public disclosure and deposit decision

The OpenAI source is cited as supplied, with a manuscript date of October 4, 2026. The repository history observed on October 6 has the pinned initial commit timestamped October 6, 21:58:50 UTC. A manuscript date and a commit timestamp alone do not establish its earliest public disclosure; this record makes no first-priority claim.

An affirmative duplication witness is the author’s publicly accessible research triage [6], in commit `f27318d83bd7000ef817957a9a4b3087de28d198`, timestamped October 7, 04:55:38 UTC (October 6, 21:55:38 PDT). Its report, Section 4, states all the present core consequences, and its algebra notes give $e = 1 - ba$, $P = eR$, the direct-sum splitting and the left-multiplication isomorphism. The triage expressly conditioned its recommendation on validating the upstream theorem. The present note adds a source audit and expanded verification; it does not add a new idempotent or cancellation mechanism or a new mathematical conclusion.

Under the project’s duplication rule, no duplicate preprint is deposited on Zenodo and no DOI or tracker entry is created. Standard further consequences of a proper one-sided inverse do not warrant relabeling the already disclosed result as a new discovery.

Disclosure. AI tools were used extensively in research, drafting and verification, including independent automated adversarial audits. This note has not undergone conventional human peer review or refereeing. OpenAI is the supplied author of the upstream AI-produced construction; no affiliation or coauthor is inferred.

References

- [1] OpenAI, *A Torsion-Free Group Algebra That Is Not Directly Finite*, OpenAI Math Release preprint, October 4, 2026, Theorem 1.1. Pinned source commit [adc7f124](#).
- [2] Johan Öinert, *Units, zero-divisors and idempotents in rings graded by torsion-free groups*, [arXiv:1904.04847v3](#), July 20, 2023 (first submitted April 9, 2019).
- [3] Giles Gardam, *The Kaplansky conjectures*, lecture notes, source version February 20, 2024, commit `ec842e6`, Proposition `kaplansky_relations`.
- [4] P. Ara, K. R. Goodearl, K. C. O’Meara and E. Pardo, *Separative cancellation for projective modules over exchange rings*, May 1996 report no. 142, p. 9, [institutional full text](#).
- [5] Charles A. Weibel, *The K-book: An Introduction to Algebraic K-theory*, AMS, 2013, Chapter II, [author’s chapter](#).
- [6] Alec Kriebel, *Ten further research paths opened by the OpenAI mathematics release*, public research triage, October 6, 2026, [report, Section 4](#); accompanying [algebra notes, Section 1](#).